

ROBE, Australia

WMO: 958160

Lat: 37.1775S

Lon: 139.8053E

Elev: 4

StdP: 101.28

Time Zone: 9.50 (AUA)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.0	3.3	0.4	3.9	6.2	1.5	4.2	6.7	12.6	13.1	11.5	13.0	1.1	50	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.5	32.4	18.7	29.7	18.3	27.2	17.7	20.5	27.4	19.5	26.4	18.6	24.7	5.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.2	21.9	17.4	12.4	21.2	16.5	11.8	20.4	59.1	27.9	55.7	26.6	52.9	25.0	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	10.1	9.1	-0.6	36.9	1.1	2.1	-1.4	38.5	-2.0	39.7	-2.7	40.9	-3.5	42.4		
			-0.8	22.3	1.1	1.3	-1.6	23.3	-2.3	24.0	-2.9	24.7	-3.7	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	18.3	18.1	17.0	14.8	12.9	10.6	10.5	11.1	12.0	13.4	15.4	16.9
	DBStd	3.91	3.44	3.01	3.38	2.70	2.39	2.19	1.97	1.98	2.26	2.76	3.28	3.21
	HDD10.0	59	0	0	0	0	4	18	18	11	6	2	0	0
	HDD18.3	1643	41	37	68	111	169	231	242	224	191	156	104	69
	CDD10.0	1603	257	227	216	145	94	37	35	45	64	108	161	213
	CDD18.3	146	40	31	26	6	1	0	0	0	3	15	24	
	CDH23.3	1601	457	287	250	62	4	0	0	1	2	62	179	297
	CDH26.7	632	215	112	97	12	1	0	0	0	16	59	121	
Wind	WSAvg	4.5	5.0	4.8	4.3	3.8	4.1	3.8	4.7	4.8	4.6	4.7	4.8	4.8
Precipitation	PrecAvg	641	19	17	30	46	69	93	107	91	62	44	32	29
	PrecMax	851	132	95	124	115	133	179	194	192	131	106	80	135
	PrecMin	360	1	0	1	7	11	16	40	32	20	7	4	3
	PrecStd	113	23	18	25	30	34	34	37	38	25	24	21	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.8	34.5	33.4	28.6	22.7	17.7	16.8	20.6	22.7	28.8	31.7	33.6
		MCWB	19.5	20.1	18.3	17.0	14.3	14.2	13.3	13.2	14.7	17.0	18.4	18.5
	2%	DB	32.6	30.1	29.4	25.1	19.7	16.2	15.4	17.3	20.1	24.6	28.2	30.3
		MCWB	18.9	18.7	18.1	15.6	14.2	13.2	12.3	12.5	13.1	15.3	17.9	18.0
	5%	DB	28.4	26.9	25.9	22.1	18.0	15.4	14.6	15.7	17.8	21.8	24.9	26.8
		MCWB	18.6	18.2	17.4	15.2	14.6	12.7	11.8	12.2	13.0	14.7	17.2	17.7
	10%	DB	24.8	23.8	22.7	19.9	16.8	14.5	13.9	14.5	16.2	18.9	21.5	23.2
		MCWB	17.8	17.5	16.8	15.0	13.9	12.1	11.4	11.6	12.6	14.1	16.0	17.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.9	21.0	21.2	18.2	17.1	15.4	14.3	14.7	15.8	18.0	20.2	21.0
		MCDB	31.6	28.7	26.9	23.4	18.7	16.1	15.9	17.0	19.1	24.4	27.7	26.5
	2%	WB	20.5	19.9	19.5	17.3	15.9	14.1	13.1	13.5	14.6	16.5	18.8	19.4
		MCDB	27.5	27.5	25.2	21.5	17.7	15.3	14.5	16.2	17.9	22.2	26.3	26.9
	5%	WB	19.3	18.9	18.3	16.5	15.1	13.2	12.4	12.6	13.7	15.3	17.5	18.2
		MCDB	25.7	25.2	23.4	20.0	17.2	14.7	13.9	15.0	16.7	20.2	23.2	24.5
	10%	WB	18.2	17.9	17.4	15.6	14.3	12.5	11.8	12.0	12.9	14.4	16.2	17.2
		MCDB	24.4	22.8	22.0	19.0	16.5	14.1	13.5	14.2	15.9	18.3	21.2	22.6
Mean Daily Temperature Range	5% DB	MDBR	10.5	9.9	10.3	10.0	7.6	7.7	6.4	6.9	8.0	9.5	9.7	10.0
		MCDBR	17.4	15.5	15.8	14.6	9.6	8.5	7.0	9.1	11.6	15.7	15.8	17.1
		MCWBR	6.8	6.4	6.6	7.2	6.3	6.4	5.1	6.0	6.8	8.2	7.3	7.1
	5% WB	MCDBR	13.5	12.6	12.2	10.5	7.8	7.5	6.0	7.9	9.6	13.1	13.9	13.4
		MCWBR	6.5	6.0	6.2	6.4	5.9	6.2	4.9	5.7	6.4	7.6	7.0	6.6
Clear-Sky Solar Irradiance	taub	0.340	0.338	0.326	0.318	0.308	0.304	0.309	0.319	0.340	0.358	0.337	0.337	
	taud	2.536	2.555	2.587	2.596	2.604	2.599	2.580	2.543	2.474	2.418	2.515	2.533	
	Ebn at Noon	993	973	946	894	843	816	833	878	915	941	990	1001	
	Edh at Noon	109	103	92	80	69	65	70	83	101	116	111	110	
All-Sky Solar Radiation	RadAvg	7.48	6.46	5.12	3.56	2.37	1.93	2.09	3.00	4.24	5.62	6.69	7.38	
	RadStd	0.36	0.31	0.21	0.23	0.13	0.12	0.11	0.15	0.35	0.37	0.28	0.41	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	-114	+136	+36

Nomenclature: See separate page